

Airtours Flight 28

Flight model: Boeing 737
Date: August 22, 1985

From: Manchester, UK
To: Corfu, Greece

Problem

The plane was about the take off. Suddenly, the crew hears a bang sound. The take off is aborted as the speed of the plane was lower than the decision speed and plane was brought to stop. The left engine is seems to have caught in flame. The plane finally comes to stop. Out of the 4 emergency exits, 2 were filled with smoke outside, 1 had a mechanical failure and was not functional. Smoke starts to enter into the cabin

Investigation

In the investigation, it was found that there was a hole under one of its wing and engine. Fire in the cabin was sever but should not catastrophic. A combustion can lies on the runway. Many such **combustion cans** are located in the engines, these are used to ignite fuel with air. It is possible that the cans were fractured and would have been blow making a hole in the engine which would have released large amount of fuel from the fuel tank from the wing. From the maintenance log it was found out that there was a combustion can fracture before which was repaired earlier which is common and needs a repair. The engine repair manual says that these cracks are to be welded manually but the can of concern had large than usual fracture which was repaired the same way. After which there was reports of lower thrust from the very same engine which can be due to combustion can fracture but the manual failed to mention it as a cause. The smoke entered the cabin and encircled the plane when the pilot tries the bring the plane to stop. This was followed by a slight cross wind which forced the smoke to encircle around the back of the plane. It was found that the smoke in the cabin from the burning of cabin seat cushion form and cushion cloth along with the overhead bin which all release poisonous gas. It was found that smokehood for passengers would have saved lives. By law the aircraft manufacturer must show that in their aircraft; passengers can be evacuated fast and safely. The law states that in all aircraft, all the passengers should be able to evacuate the plane with half of exits within 90 seconds. But these certification are conducted in ideal condition with no smoke and no anxiety among the test subjects which is different from real situation. The exits were jammed as the bulkhead wall/opening just next to the exit that connects the exit with the aisle of the plane was just small enough to fit exit of one person at a particular time. Increasing the width greatly decreased the probability of jamming.

Changes that were brought after the incident

The report advises to improve the width of the bulkhead walls and introduce strip lighting which help passengers evacuate even when there is thick smoke filled in the cabin. Psychologist also find that passengers get down the plane faster with a much assertive (forceful personality) cabin crew as they don't want you to make your own decision rather follow their instruction for which they are trained for. But smokehood were not implemented as they would posses a risk of slower evacuation.

Reference

1. An Engine That CAUGHT FIRE Mid-Flight! — Airtours Flight 28 — Mayday: Air Disaster
<https://www.youtube.com/watch?v=uH1GsMTvPFY>